

Rakey | Ruoqing Yang

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EDUCATION

- **University of Nottingham** 2023–2024
MSc, Human Computer Interaction (graduated with Distinction) *Nottingham, England*
- **Ningbo University** 2018–2022
BSc, Logistics Management *Ningbo, China*
- **University of Oxford** 2022
Winter School, Artificial Intelligence and Machine Learning *Oxford, England*

ACADEMIC PAPERS

- **R. Yang** and C. Greenhalgh. “Comparing Experience Intensity of AR and VR for Contrasting Phobia Stimuli.” *Virtual Reality*, in press, 2026.
- S. Jin, **R. Yang**, W. Tong, and L. H. Lee. “AIs or Humans in Luxury Shopping? A Design Study of VR Shopping Assistants.” *2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1155–1156, 2026.
- S. Jin, **R. Yang**, and L. H. Lee. “Who Should Serve You in Metaverse Luxury Stores? A Comparative Study of 3D Avatars, AI Agents, Webcam Staff, and Voice Assistants.” *Electronic Commerce Research and Applications*, under review, 2026.

ACADEMIC EXPERIENCE

- **Comparing VR and AR for Virtual Exposure Therapy** 2024–2025
MSc Dissertation, Individual Research · University of Nottingham *Nottingham, England*
 - Built the same exposure scenarios in both VR and AR to compare the two media across spider and contamination phobia, following a research-through-design approach.
 - Conducted a mixed-methods study (n=31) with questionnaires, behavioural metrics, and semi-structured interviews, analysing the data through two-way repeated-measures ANOVA and reflexive thematic analysis.
 - Found VR elicited higher experience intensity overall, while AR performed comparably and was more ecologically valid in the spider scenario, pointing to a staged AR-then-VR design.
- **User Perception of Different Shopping Assistants in VR Retail** 2025
Research Collaboration · University of Cambridge *Cambridge, England*
 - Contributed to the design of a study comparing 4 shopping-assistant modes (3D avatar, webcam, voice, AI) in a luxury VR retail setting, defined the experimental conditions and participant tasks.
 - Co-built the VR retail scenes in Unity and conducted the user study (n=20), running participant sessions and collecting and cleaning the multi-source perceptual and behavioural data.
 - Found 3D avatars to be the optimal assistant mode, while identifying a fundamental realism-immersion trade-off in asymmetric PC-VR interaction.

PROFESSIONAL EXPERIENCE

- **AI Video Specialist** 01/2024–07/2024
O Shaped · Internship *London, England*
 - Built AI avatar and generative-video workflows for educational content, cutting production time by 40%.
 - Iterated on the generated outputs from user feedback, improving clarity and usability under real production constraints.
 - Explored digital-human and embodied avatar formats for more present, human-like content delivery.

PROJECTS

- **One’s Own Room – AI-Customised WebXR Safe Zone** [\[Github\]](#)
 - Designed a WebXR app: a user-generated, interactive 3D room for real-time AI dialogue.
 - Engineered the system pipeline integrating WorldLabs’ custom 3D Gaussian splatting scenes, Three.js/Spark shaders, and real-time conversational voice interactions driven by ElevenLabs.
- **Virtual Exposure Therapy Application** [\[Github\]](#)
 - Built a Unity VR/AR exposure-therapy prototype, presenting spider and contamination scenarios.
 - Used passthrough to switch between AR and VR, with hand-tracking for controller-free interaction. Users trigger object state changes through touch and grab interactions.
- **AR Gallery Treasure Hunt Game** [\[Video\]](#)
 - Developed exhibition environment setups and modelled 3D content based on arts exhibit materials and venue geography.
 - Designed location-based AR for spatial exploration, achieving 90% task completion rate.

AWARDS & ACTIVITIES

Presented at **GOSIM AI Spotlight, Paris 2026** (Top 10 Frontier Creators), using AI Art to explore AI-human interaction.
Active hackathon participant – **First Prize**, Cambridge EduX Hackathon; **Best AI Use Prize**, RealityX (AI×XR) Hackathon.

SKILLS & LANGUAGES

Research Methods: Research-through-Design, Mixed Methods (Quant/Qual) **Data Analysis:** SPSS, Python
XR / Game Dev: Unity (C#), WebXR (Three.js, WebGL), Spark AR **Design:** Adobe Suite (Ps/Ai/Ae), Figma, Blender
Languages: English, Mandarin, Taiwanese Hokkien